Charotar University of Science and Technology
M.Sc (Advance Organic Chemistry) Semester- II Examination May 2011
OC -710: Physical Chemistry -II
Date: May 05, 2011 Time: 10.00 am to 10.30 am **Max. Marks: 20**

Part A

1. Define and explain the degree of polymerization. (2)

2. What is the difference between homopolymer and copolymer ? (2)

3. State Stark –Einstein Law of photochemical equivalence. (2)

4. Laser is the acronym of --- (1)

5. Maser is the acronym of --- (1)

6. At a given temperature, fugacity of a pure liquid would be: (2)
   i. Same as that of its vapour in equilibrium with it.
   ii. Greater than that of its vapour in equilibrium with it.
   iii. Lower than that of its vapour in equilibrium with it.

7. What is meant by quantum yield of a photochemical process? (2)

8. One einstein of energy is defined as : (2)
   i. energy absorbed by one mole of the substance.
   ii. energy absorbed by one gram of the substance.
   iii. energy absorbed by 100 g of the substance.
   Now write the expression for one einstein in terms of wave length of light absorbed.

9. $\log(I_o/I)$ is defined as ------------------- in Beer –Lambert law. This is related to (2)

concentration(c) of absorbing medium by the relation -----------------------------

where --

10. The standard state for a gas is such that: (2)

i. fugacity = activity =1 , ii. fugacity =1 & activity =0 ,

ii. fugacity =0 &activity =1, iv. fugacity =activity =0 .

11. NH_3 can give: (1)
   i. Laser , ii. Maser iii. None of these (i & ii) iv. Both of these (i & ii)

12. Explain with suitable example(s) how hydrogen bonding can account for the higher boiling point of a liquid (1)

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Part B

1(a).Explain fugacity. Derive an expression for fugacity in terms of free energy change. Describe its physical significance. (5)

(b). Calculate the free energy change accompanying the compression of 1 mole of a gas at $25^{o}c$ from 20 to 200 atm. The fugacities of gas are to be taken as 18 and 120 atm. respectively at pressures of 20 and 200 atm. (3)

(c).Explain in brief 'London Forces'. (2)

2(a). Express fugacity of a Vander Waal gas as a function of V, T, R and Vander Waal constants. (5)

OR

(a). Derive the basic principles of Laser in terms of Einstein transition probability.

(b).Explain the concept of activity. Derive an expression for activity in terms of change in chemical potential. (3)

(c).A system absorbs 2.0×10^{16} quanta of radiation per second. When irradiated for 15 minutes, it is found that 3.0×10^{-4} mole of the reactant has reacted. What is the quantum yield of the reaction? (2)

3(a). What is radical or vinyl polymerization? Discuss the kinetics of vinyl polymerization. (7)

OR

(a).Discuss the kinetics of polycondensation reaction catalyzed by acid.

(b). Write in short about "Ring Opening Polymerization Reaction" . (3)

4(a).Drive the Bragg equation. (5)

(b). Draw and discuss Jablonski diagram. Also discuss the various photophysical processes (5) involved.

5(a). Derive Stern –Volmer equation for quenching of fluorescence. (4)

(b). What is chemiluminescence ? Give some examples. Provide a mechanism of chemiluminescence due to interaction between aromatic anions and cations. (3)

(c). What is meant by photosensitization .Give three examples of photosensitized reactions. (3)

OR

(c). What is the energy of einstein associated with photon of wave length 2000A^o ?